

Curvature-matched localized Ricci fluctuations in a prepared Schwarzschild interior

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We report a reproducible numerical study of finite-resolution, matter-induced curvature fluctuations in a specified prepared state of a real, free, massless, minimally coupled scalar field on a fixed four-dimensional Schwarzschild interior. Proper-time, longitudinal, and angular sampling lengths are matched to fixed fractions of the local curvature length at areal radii $r/M = 0.60, 0.45, 0.30$. A joint pair-amplitude calculation retains the covariance between the stress trace and the squared field time derivative. At leading Einstein order these give the Ricci-scalar and observer-direction Ricci fluctuations without imposing spherical symmetry on the gravitational perturbation. Smooth angular heat kernels are compared with a whole-sphere control and a separately calculated frozen-product vacuum reference. For the narrower localized family and a common illustrative coupling $G\hbar/M^2 = 10^{-9}$, the Ricci-scalar standard deviation divided by the sampled nonzero background curvature scale increases from 3.5606×10^{-6} to 2.8285×10^{-5} . The factor 7.944 increase is close to the dimensional r^{-3} expectation for curvature-matched sampling. Both primary source standard deviations differ from the matched reference by less than 0.180%. Final tested cutoff/refinement changes in the primary variances are below 0.00254%, while a deliberately coarse pilot is rejected. These results characterize a state- and resolution-dependent finite-region effect; they do not establish a controlled quantum-dominated regime, a complete fluctuating metric, or a singularity endpoint.

I. INTRODUCTION AND QUESTION

Quantum matter can affect a gravitational description through its mean stress and through connected stress correlations. Semiclassical gravity and stochastic gravity organize these two types of information differently [1, 2]. A calculation of a nonzero quantum contribution is therefore not, by itself, a test that quantum dynamics has displaced a classical geometric approximation. The relevant questions include the quantity being measured, its resolution, its size relative to a nonzero classical scale, and the domain in which the approximation is controlled.

The motivating question of this project is whether quantum physics becomes essential in the regime that classical relativity describes as a black-hole singularity. No outward pressure, bounce, singularity removal, or Hawking origin of the stress is imposed as a premise. Here we address only one finite-region descendant of that question: do selected localized curvature fluctuations increase relative to a classical curvature scale when all chosen sampling lengths are matched inward? The answer is obtained for one explicit field and prepared state, not for a general quantum theory of the central structure.

Large stress fluctuations relative to a small mean do not alone demonstrate failure of semiclassical gravity [3]. Stability-based criteria instead involve the response of the geometry and, in general, intrinsic as well as matter-induced metric fluctuations [4, 5]. The present calculation does not implement those full criteria. It evaluates two local, linearly gauge-invariant Ricci projections using the leading Einstein relations, and distinguishes their numerical and scale checks from a theorem of semiclassical validity.

The noise kernel and its properties are established subjects [2]. In particular, approximate Schwarzschild noise calculations already exist for conformally invariant scalar fields in thermal states [6]. The present state is neither that thermal state nor an Unruh state, and the scalar is minimally rather than conformally coupled. The calculation uses unequal-time mode evolution, normalized spacetime filters, and exact magnetic-index sums for finite angular localization. These differences specify the computational case studied; they are not a claim that stress covariance, angular harmonic reduction, or induced Ricci fluctuations are newly invented concepts.

This paper consolidates the saved numerical release in Ref. [7]. Its contribution is a defined computational experiment, a joint covariance implementation, matched reference and resolution controls, and a reproducible account of accepted and rejected numerical settings. Earlier mean-backreaction, mass-increment, and normalized tidal-change studies in the project supply historical motivation, not interchangeable numerical evidence. Only the localized inward dataset is used for the numerical tables below. Supplemental Material [8] gives the exact preparation, all result rows, test definitions, provenance, and reproduction instructions.

II. GEOMETRY, FIELD, AND PREPARED STATE

A. Reference geometry and dimensions

We use signature $(-, +, +, +)$ and set $c = 1$. Let $M = GM_{\text{phys}}$ denote the geometric mass. Numerical coordinates and mode momenta are measured in units of M ; thus $M = 1$ in the code. On the final measurement

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region,

$$ds^2 = A(\eta)(-d\eta^2 + dx^2) + r(\eta)^2 d\Omega^2, \quad (1)$$

$$A = 2M/r - 1, \quad \frac{dr}{d\eta} = -A, \quad 0 < r < 2M. \quad (2)$$

Here η is timelike and x is the homogeneous longitudinal coordinate. The background is the original Schwarzschild solution, not an extrapolation of a previously corrected mean geometry. Its Kretschmann scalar and the curvature length used to specify sampling are

$$K_0 = \frac{48M^2}{r^6}, \quad L_K = K_0^{-1/4}. \quad (3)$$

The timelike sampling direction is $u^a = A^{-1/2}(\partial_\eta)^a$. The metric is spherically symmetric, but the angularly localized test functions below are not uniform over a sphere.

The final field action is

$$S_\phi = -\frac{1}{2} \int d^4x \sqrt{-g} g^{ab} \nabla_a \phi \nabla_b \phi. \quad (4)$$

The field is free, real, massless, and minimally coupled in the region of observation. A smooth auxiliary mass and auxiliary geometry are used only to specify its state. They are not assumed to satisfy the semiclassical Einstein equation or to describe a collapsing star.

B. State prescription and mode normalization

The state, denoted SPV-1 in the numerical record, is the vacuum obtained by evolving the ground-state modes of an ultrastatic massive past through a specified smooth transition. The dimensionless transition time is $0 \leq \eta \leq 2$ and the geometry becomes Schwarzschild at $r/M = 1.5$. The auxiliary mass-squared begins at M^{-2} and vanishes smoothly. The switching function, initial radius, and differential equations are written explicitly in Supplemental Sec. S1 and implemented in `prepared_geometry.py`. This fixes a mathematical in-state; it does not identify an astrophysical collapse vacuum.

With harmonics normalized by $\int d\Omega Y_{\ell m}^* Y_{\ell p n} = \delta_{\ell p} \delta_{mn}$, write

$$\phi = \sum_{\ell m} \int_{-\infty}^{\infty} \frac{dk}{\sqrt{2\pi}} [a_{k\ell m} f_{\ell k}(\eta) e^{ikx} Y_{\ell m}(\Omega) + \text{h.c.}], \quad (5)$$

where $[a_{k\ell m}, a_{k'p n}^\dagger] = \delta(k - k') \delta_{\ell p} \delta_{mn}$. In numerical units $\hbar = 1$; it is restored in the reported variances. Setting $\chi_{\ell k} = r f_{\ell k}$ gives

$$\chi_{\ell k}'' + \left[k^2 + A \left(\frac{\ell(\ell+1)}{r^2} + m_{\text{aux}}^2 \right) - \frac{r''}{r} \right] \chi_{\ell k} = 0, \quad (6)$$

with $\chi \chi^{*'} - \chi' \chi^* = i$. The initial modes are positive-frequency ground-state modes in the ultrastatic past. Production evolves a real width/phase representation;

separate complex-mode integration is used for representative checks. No absolute renormalized mean-stress table is needed to generate the covariance in this work.

Smooth preparation from a regular massive ground state is the intended ultraviolet state prescription. The present numerical release tests normalization and convergence but does not supply a standalone microlocal proof for its interpolated implementation. Consequently, finite-cutoff numerical checks are not treated as a proof of every Hadamard property. The usual local Wick-observable framework supplies the appropriate setting for renormalized composite operators [9].

III. FINITE-RESOLUTION OBSERVABLES

A. Local operators and connected covariance

Let $\pi = u^a \nabla_a \phi$, and let $|\nabla_\perp \phi|^2$ include the spatial longitudinal and angular derivatives in the background orthonormal frame. The three operators of interest are

$$\mathcal{T} = \pi^2 - |\nabla_\perp \phi|^2, \quad \mathcal{P} = \pi^2, \quad \rho = \mathcal{P} - \frac{1}{2} \mathcal{T}. \quad (7)$$

For the minimally coupled massless field, $\mathcal{T} = T^a_a$ and $\rho = T_{ab} u^a u^b$, up to local identity-valued renormalization terms. The quadratic operator is formed before it is averaged; this is not the square of an averaged linear field.

For a specified common filter, let Q_A be the averaged operator and $\Delta Q_A = Q_A - \langle Q_A \rangle$. The real covariance is

$$C_{AB} = \frac{1}{2} \langle \Delta Q_A \Delta Q_B + \Delta Q_B \Delta Q_A \rangle, \quad A, B \in \{\mathcal{T}, \mathcal{P}\}. \quad (8)$$

The supplementary energy variance uses the same matrix,

$$\text{Var}(Q_\rho) = C_{\mathcal{P}\mathcal{P}} + \frac{1}{4} C_{\mathcal{T}\mathcal{T}} - C_{\mathcal{T}\mathcal{P}}. \quad (9)$$

It is not obtained by adding independent energy and pressure variances. This combination can be much smaller than the parent entries and is therefore a numerically cancellation-sensitive observable.

Adding a fixed local multiple of the identity to a renormalized mean operator leaves ΔQ_A unchanged. This elementary invariance does not license a divergent pointwise product, settle contact-term questions for every composite observable, or remove the requirement of a specified state and smearing. The connected-noise framework and covariant renormalization issues are distinct from deleting the vacuum contribution [2, 10]. In this calculation finiteness is approached through positive pair norms of the explicitly sampled operators and tested spectral extensions.

B. Time, longitudinal, and angular filters

The averaged observable is

$$Q_A = \int d\tau F_\sigma(\tau) \int dx G_{b_x}(x) \int d\Omega H_\delta(\Omega) \mathcal{O}_A(\tau, x, \Omega), \quad (10)$$

Here $\mathcal{O}_A$ denotes the selected local operator. All three filters are normalized. The proper-time kernel is proportional to $e^{-(\tau/\sigma)^2}$, with a smooth compact taper equal to one through $|\tau| = 4\sigma$ and zero at 5σ . The proper-time origin is at the chosen central radius. Its exact definition is in Supplemental Sec. S2.

The longitudinal filter is

$$G_{b_x}(x) = \frac{e^{-x^2/b_x^2}}{\sqrt{\pi}b_x}, \quad b_x = \frac{b_c}{\sqrt{A_c}}. \quad (11)$$

Thus b_c is the proper exponent width at the center, not at every time in the support. It is not a standard deviation. The proper width evolves as $b_x \sqrt{A(\tau)}$. This distinction is retained in the scale screens.

The angular weight centered at a chosen pole is the positive spherical heat kernel

$$H_\delta(\theta) = \sum_{J=0}^{\infty} \frac{2J+1}{4\pi} e^{-\delta^2 J(J+1)/2} P_J(\cos \theta). \quad (12)$$

It has global tails; it is not a hard cap. The uniform control is $H = 1/(4\pi)$ and retains only $J = 0$. It is not the limit $\delta \rightarrow 0$, which localizes rather than averages. The effective angular area fraction is

$$f_{\text{eff}} = \left[\sum_{J=0}^{\infty} (2J+1) e^{-\delta^2 J(J+1)} \right]^{-1}. \quad (13)$$

Equation (10) can be written with the invariant space-time volume by dividing the scalar test weight by $\sqrt{A}r^2$. This makes clear that it is a specified spacetime observable. It is not a point detector or an assertion of independent random cells.

C. Curvature-matched inward families

At $x_c = r_c/M \in \{0.60, 0.45, 0.30\}$, set

$$\begin{aligned} s_c &= (x_c/0.60)^{3/2}, & \sigma/M &= 0.02s_c, \\ b_c/M &= 0.25s_c, & \delta &= \delta_0 \sqrt{x_c/0.60}, \end{aligned} \quad (14)$$

with $\delta_0 = 0.25$ and 0.50 . Then σ , b_c , and $r_c\delta$ are fixed fractions of $L_K(r_c)$. They do not make the full geometry, directional expansion rates, or every point in a finite window identical. A whole-sphere control is retained but is not itself a curvature-matched local angular patch.

Table I collects the sampling parameters and full time-support intervals.

TABLE I. Primary sampling families. Radii and lengths are in units of M . The last column is the entire compact time support expressed as an areal-radius interval, not only its quadrature nodes. The wider angular family has twice the listed δ .

x_c	σ/M	b_c/M	δ_{narrow}	r/M support
0.60	0.020000	0.250000	0.250000	[0.430227, 0.740781]
0.45	0.012990	0.162380	0.216506	[0.316538, 0.561633]
0.30	0.007071	0.088388	0.176777	[0.207128, 0.378262]

All supports lie above the retained analytic field-domain boundary $r/M = 0.2$. The centers are deep inside the horizon but are not arbitrarily close to $r = 0$. No early-to-late difference kernel is used here, unlike the separate mass-increment and normalized tidal-change studies in the project.

IV. PAIR-AMPLITUDE COMPUTATION AND REFERENCE

A. Angular reduction and the joint matrix

In a Gaussian vacuum, Wick contraction reduces Eq. (8) to a two-particle norm. Define $d_{\ell k} = f'_{\ell k}$ and, for two angular orders ℓ, p and two positive momenta k, k' , the time-integrated matrices

$$\begin{aligned} D_{\ell p} &= \int d\tau F_\sigma \frac{d_{\ell k} d_{pk'}}{A}, \\ X_{\ell p} &= \int d\tau F_\sigma \frac{f_{\ell k} f_{pk'}}{A}, \\ Z_{\ell p} &= \int d\tau F_\sigma \frac{f_{\ell k} f_{pk'}}{r^2}. \end{aligned} \quad (15)$$

There is no complex conjugation inside an aa amplitude. Conjugation enters when taking its norm. For the two relative momentum signs $s = \pm 1$,

$$B_J^{(s)} = D + s k k' X - \frac{\ell(\ell+1) + p(p+1) - J(J+1)}{2} Z. \quad (16)$$

The source-amplitude vector is $v_J^{(s)} = (B_J^{(s)}, D)^T$. The energy amplitude is $D - B_J^{(s)}/2$.

Using the Gaunt identity and magnetic-index orthogonality [11], the covariance spectrum in the conventions

of Eq. (5) is

$$\mathcal{G}_{\ell p J} = \frac{(2\ell+1)(2p+1)(2J+1)}{16\pi^4} \begin{pmatrix} \ell & p & J \\ 0 & 0 & 0 \end{pmatrix}^2,$$

$$S_{AB}^J = \sum_{\ell \leq p} (2 - \delta_{\ell p}) \mathcal{G}_{\ell p J} \int_0^\infty dk \int_0^\infty dk' \\ \times \sum_{s=\pm 1} e^{-b_x^2(k+sk')^2/2} \text{Re}[v_{J,A}^{(s)} v_{J,B}^{(s)*}], \quad (17)$$

$$C_{AB}(\delta) = \sum_J e^{-\delta^2 J(J+1)} S_{AB}^J. \quad (18)$$

Triangle and even-parity conditions are imposed. The factor of two from the vacuum two-particle norm, momentum-sign reduction, and unequal-angular-order multiplicity are included. The uniform case selects S^0 . Thus averaging the gravitational observable over a sphere does not mean retaining only scalar field modes with $\ell = 0$.

Each angular pair stores six real moments from which the entire 2×2 spectrum can be rebuilt for every retained J . Positive pair norms establish the expected covariance positivity; independent angular, mode-phase, and direct-pair checks test the implementation. No covariance element or pressure is adjusted to impose an identity.

B. Frozen-product control

The independent reference freezes r and A at the central values and uses a massless positive-frequency state on the resulting ultrastatic product geometry. Its mode frequencies are

$$\omega_{\ell k}^{\text{fr}} = \sqrt{k^2 + A_c \ell(\ell+1)/r_c^2}. \quad (19)$$

The same proper-time, longitudinal, and angular filters are applied. Products in Eq. (15) are replaced by their analytic frequency representation, including

$$f_\ell f_p \mapsto \frac{\hat{F}_\sigma[(\omega_\ell + \omega_p)/\sqrt{A_c}]}{2r_c^2 \sqrt{\omega_\ell \omega_p}}, \quad (20)$$

with $D = -\omega_\ell \omega_p f_\ell f_p / A_c$. The compact-kernel transform is evaluated with a separately checked interpolation against direct quadrature.

This is a derivative-stress comparison, not an Einstein solution on the frozen product and not the same global state or history as the interior. The massless undifferentiated zero-mode infrared issue is not claimed to be resolved for general ϕ^2 ; the calculation concerns derivative observables. Agreement with the reference is not an exact partition of the measured noise into independent static and dynamical contributions. In particular, variances from these two different models should not be subtracted as though they were independent random sources.

V. LEADING GRAVITATIONAL INTERPRETATION

At leading Einstein order on the Ricci-flat background,

$$\begin{aligned} \delta R &= -8\pi G \Delta \mathcal{T}, \\ \delta(R_{ab} u^a u^b) &= 8\pi G \Delta \mathcal{P}, \\ \delta(G_{ab} u^a u^b) &= 8\pi G \Delta \rho. \end{aligned} \quad (21)$$

Here Δ refers to the centered source. These are the necessary local projections of a matter-induced perturbation; they are not a reconstruction of its complete metric. Since the reference Ricci tensor is zero, its first perturbation and these specified background-frame projections are linearly gauge invariant. Frame changes multiplying the background Ricci tensor do not contribute at this order. This does not remove gauge and constraint questions for a full Weyl or metric response.

The source cross entry $C_{\mathcal{T}\mathcal{P}}$ is not the Ricci cross entry. The latter is obtained by multiplying the source matrix on both sides by $(8\pi G) \text{diag}(-1, 1)$. The energy combination is an Einstein-tensor projection, not an additional Ricci eigenvalue.

For gravitational magnitude we use a nonzero background scale,

$$D_c = \int d\tau F_\sigma(\tau) \frac{\sqrt{48}}{[r(\tau)/M]^3}, \quad \epsilon = \frac{G\hbar}{M^2}, \quad (22)$$

so that, for the appropriate numerical variance V_A ,

$$\chi_A = \frac{\text{SD}(\delta \mathcal{R}_A)}{\langle \sqrt{K_0} \rangle} = \epsilon \frac{8\pi \sqrt{V_A}}{D_c}. \quad (23)$$

Physical stress standard deviations have units $\hbar/M^4$, and their variances have units $\hbar^2/M^8$. The denominator is neither background Ricci, which vanishes, nor a nearly zero quantum mean. Nevertheless, χ_A is only one scale comparison. It is not the variance of K , a percentage of all gravity, a maximum excursion, or a rare-event probability. Quadratic-field statistics need not be Gaussian; a second moment does not supply the probability distribution [12].

VI. NUMERICAL DESIGN AND VALIDATION

The production record comprises seven campaigns: one coarse pilot and a main/fine pair at each radius. All are separately saved, not inferred from counts in a narrative. In total, 187,904 (ℓ, k) labels were evolved and 896 angular contraction blocks stored. The count is not multiplied by harmonic degeneracy or the number of pair contractions. The finest configurations are shown in Table II; full main and pilot settings are in Supplemental Table S2.

No fitted ultraviolet tail is added to the final covariance. Positive-momentum integration uses Gauss-Legendre nodes. Time integration uses the actual compact support and normalized proper-time weights. The

TABLE II. Finest production settings. L is the largest scalar angular order, $K_{\max}$ the positive longitudinal cutoff, and n_k, n_τ the quadrature sizes. All retain external harmonics through $J = 32$ and use ODE relative tolerance 2×10^{-11} .

r_c/M	L	$K_{\max}M$	n_k	n_τ
0.60	127	256	192	320
0.45	159	480	224	384
0.30	191	960	256	448

TABLE III. Largest primary-variance changes across the angular filters, expressed as percentages. These are numerical sensitivities, not statistical or physical uncertainty intervals.

r_c/M	Field L	Matched- L	$J : 24 \rightarrow 32$
0.60	6.17×10^{-6}	9.90×10^{-6}	$< 4 \times 10^{-14}$
0.45	1.86×10^{-8}	4.13×10^{-5}	1.36×10^{-11}
0.30	7.85×10^{-9}	2.538×10^{-3}	2.77×10^{-7}

ODE implementation uses DOP853 with an absolute tolerance equal to the relative tolerance divided by 100. Exact end states and source/configuration fingerprints are saved per angular order. Approximate, phase-resolved history caches were built afterward and checked separately; they are not inputs replacing the production time integrals.

The declared tests distinguish scalar-field angular cutoff L from external measurement harmonic cutoff J . For each output, the fine-grid spectra are restricted to the main L and compared with the unrestricted fine result. The restricted fine result is then compared with the main result; this second comparison changes momentum cutoff, quadrature, time resolution, and solver tolerance jointly. It does not isolate those errors one at a time. The reference is tested in the same manner.

Table III summarizes the primary numerical refinement comparisons.

All 43 Boolean entries in the archived final acceptance record are true, including supplementary energy checks, reference checks, overlap, and positive covariance. The representative independent tests include symbolic Wigner coefficients, Gaunt and angular-gradient quadratures, direct versus moment-reduced pairs, independent complex versus width/phase modes, proper-time support, reference Fourier integration, phase invariance, and trace reversal. These tests do not amount to a complete position-space Ward-identity audit of every component of a tensor noise kernel.

The pilot differs from the finest results by as much as 30.48% for a reported variance and is rejected. Its outputs remain available. The smallest normalized eigenvalue of the reported source covariance matrices is approximately 1.50×10^{-4} , consistent with positive definiteness while also showing that some linear combinations are much smaller than the parent entries. Further accu-

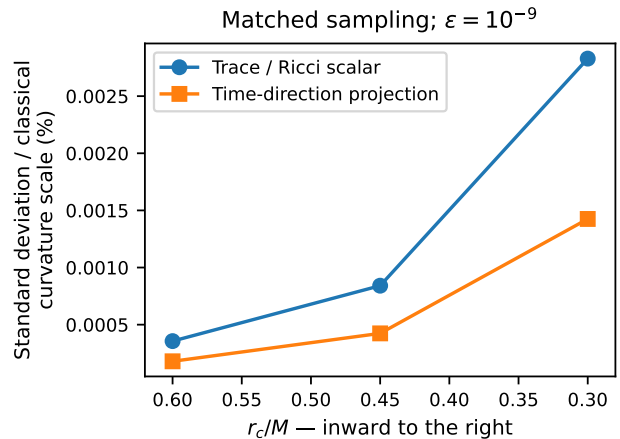


FIG. 1. Primary narrow-family induced-curvature standard deviations relative to the sampled background scale, at a common $\epsilon = 10^{-9}$. The horizontal direction runs inward. Curves connect three computed points and are not an extrapolation toward the singularity.

racy issues for such combinations must be assessed with conditioned rather than solely entrywise relative norms.

A separate manuscript-stage audit verified all 4,169 file hashes in the source release, reduced the 480 fine-grid contraction blocks directly, and reconstructed the 27 result rows. It also reran the bounded independent-check suite in a fresh temporary tree. These are fresh verification and reassembly operations, not a new production field campaign. The primary variance differences from the saved table were below 5.5×10^{-16} relative in the current audit. Supplemental Material documents the cancellation-sensitive supplementary comparison and a different reduction-order check recorded during an earlier interrupted draft. No original numerical arrays were altered.

VII. RESULTS

A. Localized inward comparison

Figure 1 displays the two primary curvature comparisons for the narrow angular family. For the narrow family $\delta_0 = 0.25$, Table IV gives the two primary source standard deviations, their frozen-reference ratios, and the dimensionless coefficients χ_A/ϵ . All normalization factors in Eq. (23) are applied after the covariance calculation.

The Ricci-scalar comparison grows by 7.94395 between $r/M = 0.60$ and 0.30. The observer-direction comparison grows similarly. This is an increase relative to the chosen classical curvature scale, not just an increase in an unnormalized stress. It remains a comparison of finite-resolution observables whose physical supports shrink inward.

TABLE IV. Primary narrow-family results. $SD(Q)$ is in units $\hbar/M^4$; reference ratios compare source standard deviations, not gravitational dominance. Percentages use the same illustrative $\epsilon = 10^{-9}$ at every radius. Extra digits facilitate replication and do not indicate a total physical error bar.

r_c/M	Projection	$SD(Q)$	$SD(Q)/SD(Q_{fr})$	D_c	χ_A/ϵ	$100\chi_A$
0.60	δR	4586.4380	0.99820350	32.374004	3560.5655	0.00035606%
0.45	δR	25700.9891	0.99821225	76.803681	8410.2259	0.00084102%
0.30	δR	291972.0135	0.99822271	259.433293	28284.9474	0.00282849%
0.60	δR_{uu}	2310.2611	1.00038081	32.374004	1793.5129	0.00017935%
0.45	δR_{uu}	12947.4773	1.00048331	76.803681	4236.8489	0.00042368%
0.30	δR_{uu}	147107.5590	1.00060752	259.433293	14251.1247	0.00142511%

The wider angular family and uniform control are also included in Supplemental Table S1. For example, at $r/M = 0.60$ the trace standard deviations for uniform, $\delta = 0.50$, and $\delta = 0.25$ are approximately 1141.23, 2377.25, and 4586.44. Narrowing the angular weight increases the sampled fluctuation. The effective area of the narrow family decreases from 6.12% to 3.09% of the sphere over the inward comparison, even though its proper diffusion-width convention is matched to curvature. The uniform control should not be confused with that matched local family.

B. What the reference does and does not explain

Figure 2 displays the departures of the two primary source standard deviations from their matched reference values. Across all nine primary windows, the trace/reference standard-deviation ratio lies between 0.9982035 and 0.9982291; the corresponding time-projection ratio is between 1.0003754 and 1.0006075. Both are within 0.180% of unity. This agreement indicates that a substantial part of the observed primary behavior is already accounted for by the reference with matched filters. It does not identify the residual as an isolated new physical contribution, nor show that preparation and geometry are individually negligible.

The reference result is itself quantum. Agreement with it therefore does not make the measured fluctuation classical, and does not logically rule out quantum importance at other scales. A residual-to-reference ratio and a ratio to classical curvature answer different questions. The former tests a comparison model; the latter assesses one gravitational magnitude at the chosen coupling.

C. Supplementary energy combination

The energy combination of Eq. (9) is retained as a secondary diagnostic. In the narrow family its source/reference RMS ratios are 2.736, 2.872, and 3.032 inward. Its standard deviations are only about 74.93, 441.93, and 5315.17 in the same field units, much smaller

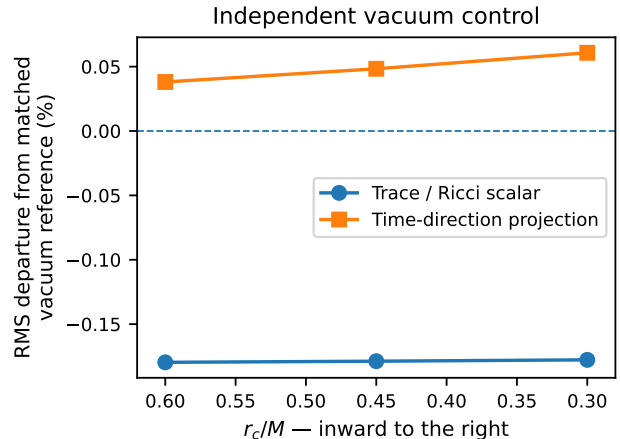


FIG. 2. Narrow-family fractional RMS departures from the independently calculated frozen-product reference. No coefficients are fitted to the interior data. The comparison is local to the specified operators and filters, not an equality of full geometries or states.

than the primary trace values. Their gravitational percentages at $\epsilon = 10^{-9}$ are approximately 0.00000582%, 0.00001446%, and 0.00005149%.

This is a difference of strongly correlated terms. Its larger ratio to a small reference is not a larger ratio to classical gravity. It cannot be interpreted as a new independent noise source. No additional prepared-state sweep was performed in the localized inward release, and the earlier project's prepared-state controls involve different observables. Their insensitivity is not transferred to the present energy diagnostic.

VIII. VALIDITY, INTERPRETATION, AND LIMITATIONS

A. A conditional dimensional control

There is a useful dimensional interpretation of the inward trend. If short-distance stress fluctuations scale ap-

proximately as the inverse fourth power of a common sampling length, and all local lengths are proportional to L_K , then

$$\text{SD}(Q) \sim \hbar L_K^{-4}, \quad \chi \sim G \hbar L_K^{-2} \propto r_c^{-3}. \quad (24)$$

The associated factor from $0.60M$ to $0.30M$ is eight. The actual narrow-family trace factor is 7.94395, corresponding to a two-point descriptive exponent 2.9899. This is a manuscript-stage dimensional control and arithmetic summary, not a newly simulated mechanism or a fitted universal law. Finite sphere curvature, anisotropic evolution, the state, and noninfinitesimal filters prevent exact self-similarity. The reference calculation, rather than dimensional reasoning alone, provides the quantitative comparison.

B. Scale separation is not a validity theorem

At a fixed numerical configuration Eq. (23) is linear in ϵ . Changing ϵ algebraically is not a new field evolution. The illustrative 10^{-9} value gives temporal exponent widths of approximately 632, 411, and 224 Planck times for the three centers. Widths are conventions of smooth filters, not physical ultraviolet cutoffs.

The release also defines conditional screens using the minimum of the time width, the longitudinal proper width over the support, the angular diffusion width over the support where present, the minimum radius, and the minimum curvature length. Requiring this minimum to exceed $N\ell_P$ gives

$$\epsilon \leq (\lambda_{\min}/N)^2, \quad (25)$$

where $\lambda_{\min}$ is dimensionless in units M . For $N = 10, 30, 100$, the largest allowed percentage among the tabulated projections is respectively 1.4242%, 0.15825%, and 0.014242%. These are per-window maxima under chosen screening rules, not the radial history of one black hole with changing coupling.

No N here is an experimentally established transition or a sufficient effective-field-theory bound. The screens omit full polarization, all local higher-curvature response, intrinsic metric fluctuations, and other field content. Setting the extrapolated primary ratio to unity would place temporal widths at approximately 0.35–1.19 Planck times. That is outside the scale-separated claim being made. It is not a numerical location of quantum takeover or a solution of the singularity.

C. Field and gravitational scope

The scalar curvature coupling matters. For a massless conformally invariant scalar field the trace of the noise kernel vanishes, despite a possible identity-valued mean trace anomaly [2]. Thus a trace-based result for minimal

coupling cannot represent all quantum matter. The time-direction Ricci source must also be rederived when the stress tensor is improved; it is not legitimate simply to reuse $\mathcal{P} = \pi^2$ for every coupling.

The most immediate additional control is therefore an improved-stress calculation at fixed final Ricci-flat geometry and fixed final Cauchy state, comparing minimal and conformal coupling while retaining a nontrace observable. A separate calculation may vary the preparation consistently with the coupling: its auxiliary geometry is not generally Ricci flat, so those two experiments are different. No such coupling sweep is reported here.

The Ricci projections in Eq. (21) do not provide localized Weyl or tidal covariance, the retarded metric propagator, gravitational initial data, or an Einstein–Langevin solution. The distinction between induced and intrinsic fluctuations must be retained in a fuller calculation [5]. A later nonspherical response study must propagate conserved source correlations with specified initial and boundary data, rather than insert unrelated random energy kicks into homogeneous mean equations.

Other restrictions are substantive: one Gaussian prepared state, no physical collapse construction, no electromagnetic or spinor fields, no interactions, a fixed nonrotating background, three locations, smooth global angular and longitudinal tails, and no final-state geometry response in the fluctuation calculation. The numerically small changes in cutoff tests do not bound these physical approximations. Numerical convergence, field-state admissibility, gravitational approximation validity, and novelty are separate questions.

IX. CONCLUSIONS

A joint covariance of two quadratic scalar-field observables was calculated on three curvature-matched, angularly localized windows in a prepared Schwarzschild interior. The leading Einstein relations assign these covariances a limited Ricci-curvature interpretation without applying spherical mass or Weyl identities to a nonspherical patch. Full trace/time cross correlations are retained, and a supplementary energy combination is evaluated from the same pairs.

For the narrow localized family, the Ricci-scalar standard deviation relative to a nonzero sampled background curvature scale increases by a factor 7.944 when the central radius halves. At a common illustrative $\epsilon = 10^{-9}$ the ratio remains below 2.83×10^{-5} . The primary source standard deviations agree with their separately calculated frozen-product controls within 0.180%. The growth is therefore not, on its own, evidence for an unexplained singularity-specific mechanism. Agreement with an ordinary quantum reference also does not prove quantum effects can never become important; the magnitude and validity questions remain separate.

The finite-region result supports neither a universal quantum-dominance claim nor a universal exclusion of

it. It supplies a reproducible numerical test and a restriction on how growth of sampled fluctuations may be interpreted. A field-coupling/nontrace control and a specified causal nonspherical gravitational observable are more discriminating next steps than repeatedly narrowing the same windows. Quantum coherence of a rotating central ring, singularity hit/bypass probabilities, and a cosmological-origin analogy are outside the present model.

COMPETING INTERESTS

The author declares no competing interests.

FUNDING

This research received no external funding.

ETHICAL APPROVAL

Not applicable. This is a theoretical and computational physics study involving no human participants, human research data, animals, or biological materials. Ethical approval was not required.

DATA AVAILABILITY

The data and software supporting this study are available upon reasonable request from the author. They are preserved in the computational reproducibility record identified in Ref. [7], which contains the source code, configurations, covariance blocks, tabulated results, accepted and rejected numerical checks, and reproduction instructions. The accompanying supplement de-

scribes that record. A public repository release is being prepared, but its location has not yet been verified for this revision; no public repository URL or DOI is claimed. Working reviewer access and the corresponding submission-form statement must be finalized before submission.

VERIFICATION SCOPE

Archive verification and a fresh reduction of saved outputs were completed during manuscript preparation. A bounded independent-check suite was also rerun. The full seven-campaign production calculation was not rerun for the manuscript. All numerical conclusions are conditional on the documented model and validated computational procedures; no independent human specialist review or experimental confirmation is claimed.

RESEARCH PROVENANCE AND AI ASSISTANCE

Abdul Badran supplied the motivating conceptual questions and project direction. Mathematical formulations, code, numerical execution, analysis, and drafting in the computational record were developed with substantial AI assistance; these are not represented as his unaided personal derivations. ChatGPT was used for this manuscript and its source/table checks; the drafting record identifies the manuscript model as GPT-6 Astra Pro. Exact historical model versions for every earlier stage have not been independently authenticated. Instructions required source-grounded claims, preserved failure records, runnable calculations, and explicit limitations. Verification undertaken is specified above and in Supplemental Material. AI tools are not authors. The author must review, revise as needed, and take responsibility for all claims and disclosures before submission. No external submission or independent peer review is asserted by this revision.

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Supplemental methods and reproducibility evidence

Curvature-matched localized Ricci fluctuations in a prepared Schwarzschild interior

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Manuscript supplement — revised 28 September 2026

This supplement specifies the numerical experiment behind the main article. The authoritative computational basis is the unmodified `BlackHoles_Localized_Inward_Recovery.zip`. It is not a new production calculation or a compilation of all earlier project variants. Equations and tables below retain the final release’s state, sampling, operator conventions, and uncertainty qualifications. The current manuscript-stage checks are identified separately in Sec. S8.

S1 Exact prepared state and field evolution

In numerical units $M = 1$, define

$$r_*(r) = r + 2 \ln(1 - r/2), \quad h(r) = 2/r - 1. \quad (\text{S1})$$

The dimensionless switching time is $T = 2$, and the initial radius r_p is the root in $(1.5, 2)$ of $r_*(r_p) = r_*(1.5) - 1$. The switch is

$$s(\eta) = \begin{cases} 0, & z \leq 0, \\ [1 + \exp(1/z - 1/(1 - z))]^{-1}, & 0 < z < 1, \\ 1, & z \geq 1, \end{cases} \quad z = \eta/2. \quad (\text{S2})$$

The auxiliary history is fixed by

$$r' = -hs, \quad A = 1 - s + sh, \quad m_{\text{aux}}^2 = 1 - s, \quad r(0) = r_p. \quad (\text{S3})$$

For $\eta < 0$, $A = 1$, $r = r_p$, and $m_{\text{aux}} = 1$. For $\eta \geq 2$, the field is massless and the background is Schwarzschild. The symmetry $s(2 - \eta) = 1 - s(\eta)$ implies $\int_0^2 s \, d\eta = 1$; consequently $r(2) = 1.5$. In the final region,

$$\eta(r) = 2 + r_*(r) - r_*(1.5). \quad (\text{S4})$$

The production code integrates the background with DOP853 and dense output, retaining an analytic relation for independent checks. The auxiliary history defines a quantum state rather than a semiclassical solution of stellar collapse. No stress computed on this auxiliary geometry is used to claim a physical collapse history.

The mode $\chi_{\ell k} = r f_{\ell k}$ has initial conditions

$$\chi(0) = \frac{1}{\sqrt{2\omega_p}}, \quad \chi'(0) = -i\omega_p \chi(0), \quad \omega_p = \sqrt{k^2 + \ell(\ell + 1)/r_p^2 + 1}. \quad (\text{S5})$$

The Wronskian is $\chi \chi^{*'} - \chi' \chi^* = i$. For the numerical width/phase method, let

$$q = k^2 + A[\ell(\ell + 1)/r^2 + m_{\text{aux}}^2], \quad w = \sqrt{q}, \quad W = w + d, \quad \chi = \frac{e^{-i\vartheta}}{\sqrt{2W}}. \quad (\text{S6})$$

Writing $\chi'/\chi = u - iW$ yields the real evolution system

$$u' = 2wd + d^2 + r''/r - u^2, \quad (\text{S7})$$

$$d' = -2u(w + d) - q'/(2w), \quad (\text{S8})$$

$$\vartheta' = w + d. \quad (\text{S9})$$

The direct complex oscillator equation is integrated independently for representative checks. Global constant mode phases cancel from the covariance. Positivity and normalization of the width representation are checked; no negative-width or failed integrator result is treated as a physical mode.

S2 Sampling functions and physical support

S2.1 Proper time

For a monotonically inward comoving trajectory, define

$$P(r) = 2 \arcsin \sqrt{r/2} - \sqrt{r(2-r)}, \quad \tau = P(r_c) - P(r). \quad (\text{S10})$$

Thus τ increases inward and $d\tau = \sqrt{A} d\eta$. The normalized time weight is

$$F_\sigma(\tau) = \frac{e^{-z^2} c(z)}{\sigma \mathcal{N}}, \quad z = \tau/\sigma, \quad \mathcal{N} = \int_{-5}^5 e^{-z^2} c(z) dz, \quad (\text{S11})$$

where

$$c(z) = \begin{cases} 1, & |z| \leq 4, \\ [1 + \exp(-1/v + 1/(1-v))]^{-1}, & 4 < |z| < 5, \quad v = |z| - 4, \\ 0, & |z| \geq 5. \end{cases} \quad (\text{S12})$$

Gauss–Legendre quadrature on $z \in [-5, 5]$ is multiplied by $e^{-z^2} c(z)$ and normalized. The complete endpoint support is checked through $P(r)$, not inferred from the first and last quadrature nodes. The numerical field background extends to $r/M = 0.2$; the narrowest inner radius in any time support is $0.2071283198M$. Longitudinal Gaussian and angular heat-kernel tails are not compact, so the complete spacetime test function is not a compact spatial cell.

S2.2 Longitudinal and angular normalization

The longitudinal Gaussian is $G(x) = e^{-x^2/b_x^2}/(\sqrt{\pi}b_x)$ with $b_x = b_c/\sqrt{A_c}$. Its transform is $\widehat{G}(q) = e^{-b_x^2 q^2/4}$. The squared pair-amplitude weight is therefore $e^{-b_x^2 q^2/2}$, not $e^{-b_x^2 q^2/4}$. The two real-momentum sectors have $q = k + k'$ and $q = k - k'$.

The angular weight is a normalized spherical heat kernel. With the pole along the z axis,

$$H_\delta(\Omega) = \sum_J \sqrt{\frac{2J+1}{4\pi}} e^{-\delta^2 J(J+1)/2} Y_{J0}(\Omega). \quad (\text{S13})$$

Its integral is one. Its squared L^2 norm is $\sum_J (2J+1) e^{-\delta^2 J(J+1)}/(4\pi)$, giving the effective-area convention in the main article. Rotating the pole does not change the covariance for the isotropic state and background. A finite $J_{\max}$ is a numerical truncation of a smooth kernel, not a replacement by a sharp cap.

The ratios fixed at the window center are

$$\sigma/L_K = 0.1132696163, \quad b_c/L_K = 1.4158702040, \quad r_c\delta/L_K = 0.8495221224 \text{ or } 1.6990442447. \quad (\text{S14})$$

These center ratios do not make directional expansion rates identical, and they do not make each point in a finite window locally static. The uniform control has no finite localized angular width.

S3 Angular reduction and positive pair norms

Use orthonormal complex scalar harmonics and the field-expansion normalization in the main article. The relevant Gaunt integral is

$$\int d\Omega Y_{\ell m} Y_{pn} Y_{J0} = \sqrt{\frac{(2\ell+1)(2p+1)(2J+1)}{4\pi}} \begin{pmatrix} \ell & p & J \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \ell & p & J \\ m & n & 0 \end{pmatrix}. \quad (\text{S15})$$

Integration by parts on the unit sphere gives

$$\int d\Omega Y_{J0} \nabla Y_{\ell m} \cdot \nabla Y_{pn} = \frac{\ell(\ell+1) + p(p+1) - J(J+1)}{2} \int d\Omega Y_{J0} Y_{\ell m} Y_{pn}. \quad (\text{S16})$$

Magnetic-index orthogonality removes cross terms between distinct J . The surviving $3j$ symbols require $|\ell - p| \leq J \leq \ell + p$ and even $\ell + p + J$. Summing field pairs only for $\ell \leq p$ adds a factor $2 - \delta_{\ell p}$. These identities and exact-symbolic versus numerical weights are checked independently; standard harmonic identities are not claimed as new results [main article, Ref. DLMF].

For a normal-ordered quadratic operator Q , the part of $Q|0\rangle$ orthogonal to the vacuum consists of two-particle amplitudes. The connected covariance is their real inner product. This gives a positive semidefinite Gram matrix and explains why both source projections must use the same field pairs. Additive identity-valued terms drop out after centering, but they do not turn an unsmeared product into a well-defined observable. The prescribed spacetime filters and high-frequency convergence remain essential.

For an angular pair, set $L = \ell(\ell + 1)$, $P = p(p + 1)$ and define D, X, Z as in the main article. For a momentum sign s , let

$$M_0 = D - \frac{1}{2}(L + P)Z, \quad B_0 = M_0 + skk'X, \quad a_J = \frac{1}{2}J(J + 1). \quad (\text{S17})$$

The six stored real moments, including momentum quadrature and the longitudinal factor, are

$$v_0 = \langle |B_0|^2 \rangle, \quad v_1 = \langle \text{Re}(B_0 Z^*) \rangle, \quad v_2 = \langle |Z|^2 \rangle, \quad (\text{S18})$$

$$v_3 = \langle |D|^2 \rangle, \quad v_4 = \langle \text{Re}(B_0 D^*) \rangle, \quad v_5 = \langle \text{Re}(Z D^*) \rangle. \quad (\text{S19})$$

Brackets here mean the numerical k, k', s integral, not a quantum expectation value. Before the angular coefficient, the joint matrix is

$$\begin{pmatrix} v_0 + 2a_J v_1 + a_J^2 v_2 & v_4 + a_J v_5 \\ v_4 + a_J v_5 & v_3 \end{pmatrix}. \quad (\text{S20})$$

Summing with the factor in the covariance-spectrum equation of the main article and the heat-kernel coefficient recovers the full covariance. The independent table audit uses this reduction on saved blocks rather than copying the final CSV.

S4 Reference calculation and gravitational projection

S4.1 Reference is a control, not a replacement source

At fixed r_c, A_c , define $\omega_{\ell k} = \sqrt{k^2 + A_c \ell(\ell + 1)/r_c^2}$. A mode pair carries proper-time frequency $(\omega_{\ell k} + \omega_{pk'})/\sqrt{A_c}$. The same normalized time kernel is transformed at this frequency, and the same longitudinal and angular kernels are used. Multiplication by $-\omega_{\ell} \omega_p / A_c$, $1/A_c$, or $1/r_c^2$ gives D , X , or Z , respectively, from the common product.

The compact-kernel Fourier transform is evaluated on a dense grid and interpolated. Direct transform and time-quadrature tests check that acceleration. Reference cutoffs are varied independently. The local reference is not fitted to the numerical interior. It uses another geometry and state; close agreement does not establish an exact additive separation into static noise and a dynamical excess. The derivative observables remain distinct from an undifferentiated massless $\ell = 0$ vacuum two-point function with its infrared qualification.

S4.2 Units and nonzero normalization

Numerical source variances have units $\hbar^2/M^8$. Let D_c denote the sampled dimensionless background quantity

$$D_c = \int d\tau F_\sigma(\tau) \frac{\sqrt{48}}{r(\tau)^3}. \quad (\text{S21})$$

The values are 32.37400395, 76.80368124, and 259.43329339 for $r_c/M = 0.60, 0.45, 0.30$. The ratio for a source variance C_{AA} is

$$\chi_A = \epsilon \frac{8\pi\sqrt{C_{AA}}}{D_c}, \quad \epsilon = G\hbar/M^2. \quad (\text{S22})$$

The covariance of the two Ricci projections is $(8\pi G)^2 \text{diag}(-1, 1)C \text{diag}(-1, 1)$; its cross entry has the opposite sign from the source trace/time covariance. The energy combination gives the observer-direction Einstein projection. No ratio is taken to the vanishing background Ricci scalar or to a vanishing quantum mean.

The equations are leading Einstein-order local projections on a Ricci-flat reference. They do not determine the full constrained, retarded metric solution, Weyl curvature, intrinsic gravitational fluctuations, or higher-order polarization. The frozen reference is not put into those equations as a separate self-consistent gravitational spacetime.

S5 Complete tabulated result

Table S1 contains all 27 projection/window rows. “U” is uniform; “W” and “N” are wider and narrower heat-kernel families. The operators are $T = \mathcal{T}$, $P = \mathcal{P}$, and $E = \rho$. The source standard deviation is in $\hbar/M^4$. The final column is $100\chi_A$ for the illustrative common $\epsilon = 10^{-9}$. Full precision and exact covariance spectra are supplied as CSV, not inferred from printed rounded values.

Table S1: Complete result rows from the saved numerical release.

r_c/M	Operator	Window	f_{eff}	Source SD	SD/reference	$100\chi_A$ (%)
0.60	T	U	1.00000	1141.2	0.9982070	0.0000886
0.60	T	W	0.22984	2377.2	0.9982063	0.0001846
0.60	T	N	0.06121	4586.4	0.9982035	0.0003561
0.60	P	U	1.00000	572.66	1.0003785	0.0000445
0.60	P	W	0.22984	1194	1.0003754	0.0000927
0.60	P	N	0.06121	2310.3	1.0003808	0.0001794
0.60	E	U	1.00000	17.646	7.0657447	0.0000014
0.60	E	W	0.22984	36.806	5.4041537	0.0000029
0.60	E	N	0.06121	74.933	2.7359660	0.0000058
0.45	T	U	1.00000	5553	0.9982172	0.0001817
0.45	T	W	0.17607	13216	0.9982161	0.0004325
0.45	T	N	0.04615	25701	0.9982122	0.0008410
0.45	P	U	1.00000	2786.7	1.0004775	0.0000912
0.45	P	W	0.17607	6638.2	1.0004786	0.0002172
0.45	P	N	0.04615	12947	1.0004833	0.0004237
0.45	E	U	1.00000	90.579	7.4597071	0.0000030
0.45	E	W	0.17607	216.68	5.6947923	0.0000071
0.45	E	N	0.04615	441.93	2.8718573	0.0000145
0.30	T	U	1.00000	51644	0.9982291	0.0005003
0.30	T	W	0.11988	1.4895e+05	0.9982277	0.0014430
0.30	T	N	0.03093	2.9197e+05	0.9982227	0.0028285
0.30	P	U	1.00000	25920	1.0006013	0.0002511
0.30	P	W	0.11988	74828	1.0006031	0.0007249
0.30	P	N	0.03093	1.4711e+05	1.0006075	0.0014251
0.30	E	U	1.00000	897.2	7.9507269	0.0000087
0.30	E	W	0.11988	2600.7	6.0352638	0.0000252
0.30	E	N	0.03093	5315.2	3.0324507	0.0000515

The stronger apparent excess of E relative to its small frozen reference is not a larger effect relative to classical curvature. It is a cancellation-sensitive combination of the same correlated pair amplitudes. Neither an RMS nor a covariance matrix specifies the full probability distribution. In particular, no Gaussian rare-event tail or maximum fluctuation is inferred; the stress-statistics literature cited in the main article addresses distinct sampling/state settings.

S6 Production configurations, acceptance, and failures

Seven archived campaigns contain 187,904 newly evolved (ℓ, k) labels and 896 saved correlation blocks. Counts are not multiplied by magnetic degeneracy or by the number of quadratic pairs. The three fine campaigns contain 480 blocks used by the independent manuscript-stage reduction. Their histories were generated for the stated prepared state and sampling times; prior mean-stress values cannot replace these unequal-time correlations.

All field-angular orders run from zero through the listed maximum. $J_{\text{max}} = 32$ for the angular window; comparison with $J_{\text{max}} = 24$ tests its tail. The final three campaigns use relative integrator tolerance 2×10^{-11} , versus 2×10^{-10} for the pilot/main cases. The exact absolute tolerances, momentum segmentation, solver settings, and source fingerprints are in the seven input JSON files. There is no fitted ultraviolet remainder in this calculation.

Table S2 lists the numerical settings for every production campaign.

Table S2. Production configurations.

Campaign	ℓ_{max}	$k_{max}M$	Momentum nodes	Time nodes
Pilot at 0.60M	31	112	80	128
0.60M main	95	224	160	256
0.60M fine	127	256	192	320
0.45M main	127	400	192	320
0.45M fine	159	480	224	384
0.30M main	159	800	224	384
0.30M fine	191	960	256	448

S6.1 Frozen acceptance plan

The source’s validation plan predates the new production analysis. It requires: variance and normalized matrix changes below 1% under last field-angular and matched-angular momentum/time/tolerance comparisons; external J tail below 0.1%; previous outer-window overlap within 1%; representative independent identities/residuals below 10^{-8} ; covariance positive semidefinite up to 10^{-12} relative roundoff; and successful hash/restart/corruption rejection. The supplementary energy projection is checked separately. A supplementary failure is not repaired by overwriting a pressure or covariance component.

The authoritative JSON contains 43 true gates. The largest final primary variance change is 2.5373856×10^{-5} as a fraction, or 0.0025373856%. At the outer overlap the trace/time variance discrepancies are approximately 1.74×10^{-11} and 1.06×10^{-10} . The minimum reported normalized covariance eigenvalue is 1.5033×10^{-4} , not a negative-roundoff case. These are tested numerical comparisons, not rigorous infinite-mode remainder bounds or total physical confidence intervals.

S6.2 Failures retained

The pilot’s largest variance discrepancy is 0.3047984 as a fraction and is rejected. This is not the discrepancy of the accepted fine grid. Earlier interrupted work is preserved as historical reporting, not used as a substitute for the available arrays. Small implementation or output-handling failures recorded in the original release are not evidence of a physical instability. The portable histories have their own interpolation checks; they are not bit-for-bit replacements for dense time arrays or unrestricted new-window data.

S7 Reproducibility and exact file provenance

The accompanying computational archive has 104,969,383 bytes and 4,170 entries. Its SHA-256 is

6dfce42994b5b82c15992430fc8c52336efdc
907204f5d6d59579940d453b378

The two lines concatenate to one digest. The archive’s 4,169-entry `FILE_HASHES.json` was verified during this manuscript preparation. File names and identifiers are provenance references, not public repository accessions.

These sizes and checksums identify the original computational release. The separately prepared public distribution excludes internal submission material and carries its own distribution manifest. Its retained numerical programs and scientific array files are unchanged, but that curated distribution is not byte-identical to the original ZIP. The original record remains the provenance reference for the results reported here.

Key contents are the exact state/mode/pair code; seven configurations; per-block spectra and final numerical states; 480 checked phase-history blocks; `INWARD_LOCALIZED_RESULTS.csv`; `ANGULAR_SPECTRA.csv`; `FINAL_VALIDATION.json`; the conditional scale tables; independent tests; failed outputs; and `RUN_RECORD.json`. Some inherited code supports preceding mean studies but is not invoked to calculate these covariances. Its presence is not a claim of another mean run.

After extracting the numerical archive, use a Python environment meeting its included requirements and run from its root:

```
python reproduce.py verify
python reproduce.py checks
python reproduce.py assemble
python reproduce.py full --workdir /path/to/NEW_EMPTY_DIRECTORY
```

`verify` checks bytes only. `checks` runs bounded independent tests in a fresh temporary directory. `assemble` reduces saved results and can write analysis outputs; preserve an immutable original before doing so. `full` evolves all seven campaigns in a new directory, then assembles their results. A successful process exit is not a substitute for inspecting the validation JSON. These four modes must not be reported as the same level of reproduction.

The report package also contains `audit/audit_tables.py`, which accepts the extracted archive root and an output directory. It independently reduces saved fine-block moments and reconstructs the report tables. It does not repeat a quantum-mode evolution. No external publication or data repository has been created by generating this package. Before submission, supply working reviewer access and settle a release license and persistent data/software location.

S8 Manuscript-stage verification and conditioned arithmetic

The report was prepared after an interrupted drafting attempt. A Jira progress note described earlier unshipped drafts; they were not mounted in the resumed workspace and were not treated as delivered documents. The current manuscript and supplement were rebuilt from the actual hashed numerical archive.

New bounded work in this resumed drafting turn consists of: source-manifest verification; the independent reduction of 480 fine blocks; reconstruction of all 27 printed rows; direct calculation of the inward trace ratio and descriptive two-point exponent; rerunning the independent-check suite in a fresh temporary tree; generating figures from CSV; and compiling/rendering the report sources. There was **no full production quantum-mode campaign** during report writing.

The independent reduction reproduces the angular spectra exactly. The largest relative primary-variance difference is 5.4142×10^{-16} . The supplementary energy difference is 7.9349×10^{-13} relative to its own smaller variance, and the maximum discrepancy normalized to its parent covariance scale is 5.4142×10^{-16} . The current blanket 10^{-11} relative comparison therefore passes. A prior failed drafting attempt recorded a different reduction's 3.60×10^{-11} energy-relative difference and a blanket-gate failure, with a 5.42×10^{-16} parent-scale residual. That older failure is retained as historical information; it is not presented as a new failure or silently substituted for this run.

This distinction illustrates numerical conditioning: subtracting correlated large terms can magnify relative error measured against a small residual. Both the residual-relative comparison and the parent-scale comparison should be reported. Neither justifies altering production data or relaxing a physical acceptance threshold after seeing a preferred sign. The exact current outputs and environment are in `audit/TABLE_AUDIT.json` and the associated logs.

S9 Interpretation boundaries and next discriminating test

The increase in the narrow-family trace ratio is 7.943948 as the center radius halves. A conditional short-distance scaling $SD(T) \sim \hbar L_K^{-4}$ and $\sqrt{K} \sim L_K^{-2}$ would give a ratio proportional to r^{-3} and a factor eight. This is a manuscript-stage dimensional explanation, not an independent production simulation, a proof of universality, or an exact extraction of a curvature-only signal. The matched reference supplies the quantitative check.

The principal result is for minimal coupling. An improved stress tensor at another curvature coupling changes both the trace and the nontrace source. The known conformal massless trace-noise null result is a useful control, but does not mean all stress fluctuations vanish. At fixed final Ricci-flat geometry a fixed-state operator comparison is distinct from re-preparing a state on the non-Ricci-flat auxiliary history. The planned test should separate these changes rather than combine them.

Next test field-coupling/nontrace robustness, then a causal nonspherical gravitational observable and a separately justified collapse state. None was calculated here. Ring coherence, hit/bypass probabilities, an endpoint, universal quantum dominance, and novelty remain unestablished.